Tourism digital twin and analytics lake

Estimating hotel demand from flight capacity

AeroLodge Metrics



Meet the Team

ChallengeON DCT Abu Dhabi Hackathon · NYUAD team



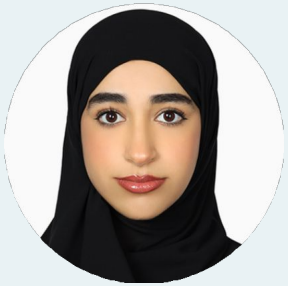
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Assumptions and questions

Can we predict the number of hotel guests from flight data?

Time Trend Analysis

01

Assumption: Air visitors grow over time, and hotel guests grow with them.

Question: How much of the change in guests and passengers is explained by time alone?

Nationality Analysis

02

Assumption: Each nationality responds differently to the same flights.

Question: Which markets bring the most guests, and does their response to flights differ?

Load Factor Analysis

03

Assumption: Fuller flights bring more hotel guests than seats alone suggest.

Question: When seats change, how much do passengers and guests really move?

Prediction

04

Assumption: Time, flights and nationality together explain most hotel guests.

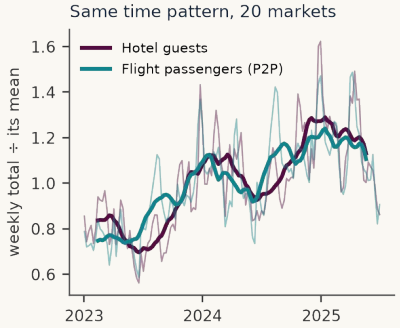
Question: How accurately can they predict guests on dates the model has not seen?

Insights and findings

Time trends explain guest and passenger demand

Time trends explain about 88% of weekly hotel guests and 87% of flight passengers.
Their time patterns move together roughly with correlation 0.62.
Growth is not linear: guests rose 30% in 2024 and 8% in 2025.

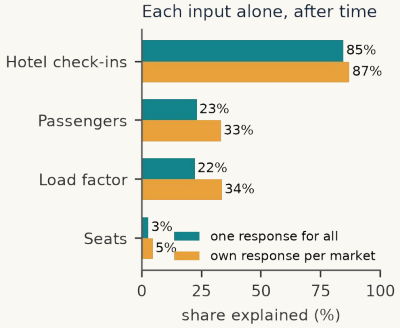
What time cannot explain is within ±20% in 84% of weeks



What time trends leave unexplained

Of the variation time does not explain, each input alone explains: hotel check-ins 85%, passengers 23%, load factor 22%, seats 3%.

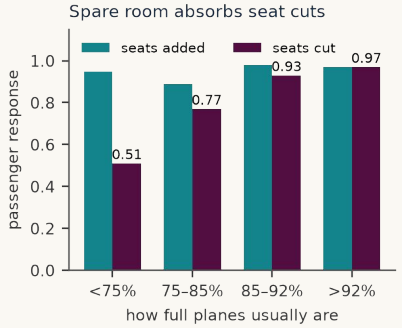
A separate response per nationality market adds about 10 points. Seats carry the trend.



Load factor absorbs change

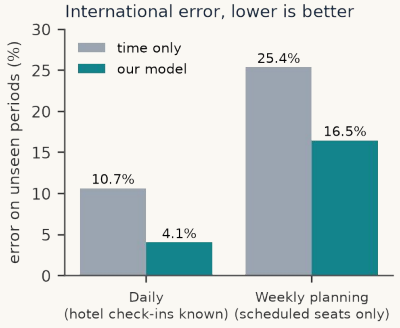
Against each market's own trend, passengers follow added seats almost one for one.

When seats are cut, planes with spare room absorb half the cut (0.51); planes over 92% full lose almost all of it (0.97).



Prediction

Hotel check-ins plus the calendar predict daily guests within 4.1% (international) and 4.2% (domestic), against 10.7% and 10.1% from time alone.



Preparing the data

Collect

Hotels: per nationality per day, guests staying (“guests”) and new arrivals (“check-ins”)

Flights: per route, airline and day, seats and passengers

Daily flights from 2023

Normalize

One key for both: market × day

Markets: the 15 largest nationalities, 5 regional groups for the rest, and Domestic

Flights join by departure country, an approximation of nationality

Clean and add history

Small hidden counts interpolated.

Each day gets its last 21 days of check-ins and last 7 days of seats

2022 left out of training

The Machine Learning Model

Two models train on the prepared data using penalised least squares with backfitting.

116,395
daily flight records

26,727
market-days used for training

45 → 21
nationalities grouped into markets

33
flight departure countries

Our solution

Two models use the same prepared data. Each fits log guest counts by least squares, updating one part at a time until the fit settles (backfitting).

Daily model
Nowcast

Hotel check-ins
Last 21 days → **Learned lag weights** → **Calendar** → **Guests today**
4.1% error, international

Planning model
Seats only

Scheduled seats
Last 7 days → **Seats to check-ins** → **Check-ins to guests** → **Weekly guests**
19.0% error

Schedule response comes from the data

Each market's response comes from the fitted chain, with a range from refits on earlier data.

Future capacity growth is an assumption

The data cannot tell whether seats drew guests or guests drew seats, so growth uses a range.

Results from flight scenarios

Estimated change per week in winter, with other settings unchanged.

UK: two more flights

2 flights per week, 290 seats each

+227

guest-nights per week

+49 hotel check-ins

+1.3% guest-nights

UK: fuller flights

Flights fill from 91% to 93%

+364

guest-nights per week

+78 hotel check-ins

2 percentage points fuller

Sweden: a new route

2 flights per week, 180 seats each

+553

guest-nights per week

+114 hotel check-ins

398 to 708 guest-nights

Winter 2026/27 outlook

Compared with winter 2025/26

+0.2%

if arrivals stay flat

+7.7%

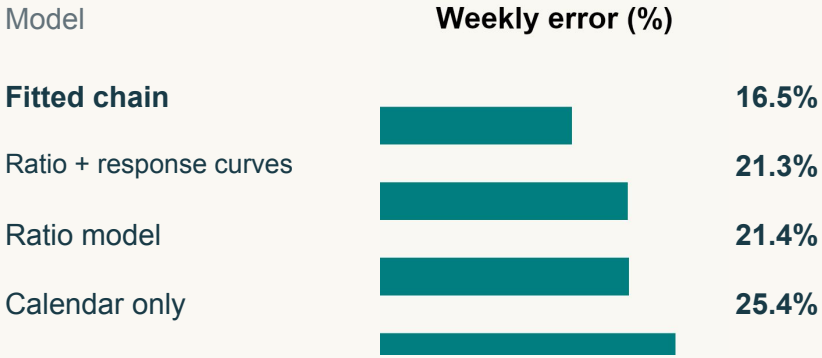
if recent growth continues

Validation against history

Daily hotel estimates and flight scenarios use different inputs and tests. Lower error is better.

Flight response benchmark

265 ordinary market-weeks in the research test



20.5% Weekly flight scenario error
30 holdout weeks, 21 markets

Daily hotel guest estimates

Daily totals, seven validation periods in 2024

Model	Domestic	International
Check-ins + calendar	**4.2%**	**4.1%**
Check-ins only	9.6%	5.4%
Calendar only	10.1%	10.7%
Same weekday last year	16.9%	23.0%

Daily estimates use known hotel check-ins.
Flight scenarios start from flight capacity.

Limits and next steps

Hotel estimates depend on how passengers reach Abu Dhabi and where they stay.

Current limits

Flight origin is a proxy

Departure country differs from nationality. Persistent gaps include the Philippines (−16%) and China (−12%).

Planning estimates have error

Weekly error is 20.5%. Seats explain about 3% of the variation left after time trends.

Room occupancy needs more data

We estimate hotel guests and guest-nights. Occupancy rates also need room inventory, which we do not have.

What we will test next

Airline and route detail

Use route and airline data to account for hub connections and differences in passenger mix.

Flights and aircraft size separately

UK test: 10% more seats moved passengers by +7.3% with extra flights, versus +5.0% with bigger aircraft.

Market relationships over time

Test whether the flight-to-hotel relationship changes by season and how hotel stays change over time.